GENETIC ENGINEERING: GM CROPS AND CRISPR

BREEDING AND ENGINEERING TAXONOMY → BT COTTON MECHANISM AND → DMH-11 CONSTRUCT AND STATUS → CRISPR TARGETING RAIL → EDITING-OUTCOME MATRIX → SDN CLASSIFICATION LADDER

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

◆ PRIMARY CORE FLOW ◇ SUBORDINATE EXTRA ◆ APPROVAL / REVIEW

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles

Approval: FALSE • Pending user review • active generation g1 • explicit approval of this exact generation is required • all prior artifacts unchanged

00

STAGE 00

BREEDING AND ENGINEERING TAXONOMY

BREEDING AND ENGINEERING TAXONOMY · CONVENTIONAL CROSS · PARENTAL GENOMES RECOMBINE SEXUALLY · OFFSPRING OF SELECTED PARENTAL LINES · SEXUALLY INCOMPATIBLE DONOR DNA · INTACT COMPATIBLE-POOL GENE · REARRANGED COMPATIBLE-POOL ELEMENTS

CONVENTIONAL CROSS → parental genomes recombine sexually

HYBRID → offspring of selected parental lines → TRANSGENIC → sexually incompatible donor DNA → CISGENIC → intact compatible-pool gene

CONCEPT / ARCHITECTURE

- CONVENTIONAL CROSS → parental genomes recombine sexually
- HYBRID → offspring of selected parental lines

MECHANISM / APPLICATION

- TRANSGENIC → sexually incompatible donor DNA
- CISGENIC → intact compatible-pool gene

READINESS / STATUS LIMIT

- INTRAGENIC → rearranged compatible-pool elements
- CONVENTIONAL CROSS → parental genomes recombine sexually

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CISGENIC → intact compatible-pool gene; INTRAGENIC → rearranged compatible-pool elements.

01

STAGE 01

BT COTTON MECHANISM AND STEWARDSHIP

BT COTTON MECHANISM AND STEWARDSHIP · CRY GENE · PLANT EXPRESSES BT PROTEIN · SPECIFIC BOLLWORM FEEDING · TOXIC ACTION IN SUSCEPTIBLE LARVAE · SELECTION PRESSURE · RESISTANT ALLELES CAN INCREASE · REFUGE + STACKING

CRY GENE → plant expresses Bt protein → SPECIFIC BOLLWORM FEEDING → toxic action in susceptible larvae → SELECTION PRESSURE → resistant alleles can increase → REFUGE + STACKING → slow resistance evolution → TRAP → insect resistance is not herbicide tolerance

STARTING CONDITION / INPUT

- CRY GENE → plant expresses Bt protein
- SPECIFIC BOLLWORM FEEDING → toxic action in susceptible larvae

SCIENTIFIC OR ENGINEERING MECHANISM

- SELECTION PRESSURE → resistant alleles can increase
- REFUGE + STACKING → slow resistance evolution

OUTPUT / FAILURE BOUNDARY

- TRAP → insect resistance is not herbicide tolerance
- CRY GENE → plant expresses Bt protein

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: TRAP → insect resistance is not herbicide tolerance.

02

STAGE 02

DMH-11 CONSTRUCT AND STATUS

DMH-11 CONSTRUCT AND STATUS · MALE STERILITY IN ONE PARENT · FERTILITY RESTORATION IN HYBRID · SELECTABLE MARKER · ENABLE HYBRIDISATION IN MUSTARD · NO COMMERCIAL-CULTIVATION OR SETTLED-LEGALITY CLAIM

BARNASE → male sterility in one parent

BARSTAR → fertility restoration in hybrid → BAR → selectable marker → PURPOSE → enable hybridisation in mustard

CONCEPT / ARCHITECTURE

- BARNASE → male sterility in one parent
- BARSTAR → fertility restoration in hybrid

MECHANISM / APPLICATION

- BAR → selectable marker
- PURPOSE → enable hybridisation in mustard

READINESS / STATUS LIMIT

- STATUS → no commercial-cultivation or settled-legality claim
- BARNASE → male sterility in one parent

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: PURPOSE → enable hybridisation in mustard; STATUS → no commercial-cultivation or settled-legality claim.

03

STAGE 03

CRISPR TARGETING RAIL

CRISPR TARGETING RAIL · GUIDE RNA · SEQUENCE RECOGNITION · TARGET DNA + PAM · PERMITTED BINDING CONTEXT · DOUBLE-STRAND CLEAVAGE · CELLULAR REPAIR · CREATES FINAL EDIT

GUIDE RNA → sequence recognition → TARGET DNA + PAM → permitted binding context → CAS9 → double-strand cleavage → CELLULAR REPAIR → creates final edit → RULE → targeting tool and repair outcome are distinct

STARTING CONDITION / INPUT

- GUIDE RNA → sequence recognition
- TARGET DNA + PAM → permitted binding context

SCIENTIFIC OR ENGINEERING MECHANISM

- CAS9 → double-strand cleavage
- CELLULAR REPAIR → creates final edit

OUTPUT / FAILURE BOUNDARY

- RULE → targeting tool and repair outcome are distinct
- GUIDE RNA → sequence recognition

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → targeting tool and repair outcome are distinct.

04

STAGE 04

EDITING-OUTCOME MATRIX

EDITING-OUTCOME MATRIX · TEMPLATE-FREE INDELS AND KNOCKOUT · DONOR-TEMPLATE SUBSTITUTION OR INSERTION · BASE EDITING · CHEMICAL BASE CONVERSION WITHOUT DSB · PRIME EDITING · TEMPLATED SHORT WRITING WITHOUT DSB · PRECISION MECHANISMS ARE NOT INTERCHANGEABLE

SYSTEM / DEVICE / INSTITUTION	DECISIVE TECHNICAL DISTINCTION	STATUS / UNIT / EVIDENCE LIMIT
NHEJ → template-free indels and knockout	BASE EDITING → chemical base conversion without DSB	TRAP → precision mechanisms are not interchangeable
HDR → donor-template substitution or insertion	PRIME EDITING → templated short writing without DSB	NHEJ → template-free indels and knockout

SYSTEM / DEVICE / INSTITUTION

- NHEJ → template-free indels and knockout
- HDR → donor-template substitution or insertion

DECISIVE TECHNICAL DISTINCTION

- BASE EDITING → chemical base conversion without DSB
- PRIME EDITING → templated short writing without DSB

STATUS / UNIT / EVIDENCE LIMIT

- TRAP → precision mechanisms are not interchangeable
- NHEJ → template-free indels and knockout

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: PRIME EDITING → templated short writing without DSB.

05

STAGE 05

SDN CLASSIFICATION LADDER

SDN CLASSIFICATION LADDER · NO TEMPLATE; SMALL EDIT · SHORT TEMPLATE; DEFINED SMALL EDIT · GENE-SIZED TARGETED INSERTION · 2022 LIGHTER TRACK · SDN-1/2 FREE OF EXOGENOUS DNA · FULL SCRUTINY · SDN-3 REMAINS OUTSIDE EXEMPTION

SDN-1 → no template; small edit

SDN-2 → short template; defined small edit → SDN-3 → gene-sized targeted insertion → 2022 LIGHTER TRACK → SDN-1/2 free of exogenous DNA

STARTING CONDITION / INPUT

- SDN-1 → no template; small edit
- SDN-2 → short template; defined small edit

SCIENTIFIC OR ENGINEERING MECHANISM

- SDN-3 → gene-sized targeted insertion
- 2022 LIGHTER TRACK → SDN-1/2 free of exogenous DNA

OUTPUT / FAILURE BOUNDARY

- FULL SCRUTINY → SDN-3 remains outside exemption
- SDN-1 → no template; small edit

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: 2022 LIGHTER TRACK → SDN-1/2 free of exogenous DNA.

06

STAGE 06

SIX-TIER BIOSAFETY INSTITUTION MAP

SIX-TIER BIOSAFETY INSTITUTION MAP · RECOMBINANT-DNA ADVICE · INSTITUTIONAL DAY-TO-DAY OVERSIGHT · RESEARCH AND CONTAINED STAGES · ENVIRONMENTAL RELEASE AND LARGE-SCALE USE · SBCC + DLC · STATE AND DISTRICT MONITORING

RDAC → recombinant-DNA advice

IBSC → institutional day-to-day oversight → RCGM / DBT → research and contained stages → GEAC / MoEF&CC → environmental release and large-scale use

CONCEPT / ARCHITECTURE

- RDAC → recombinant-DNA advice
- IBSC → institutional day-to-day oversight

MECHANISM / APPLICATION

- RCGM / DBT → research and contained stages
- GEAC / MoEF&CC → environmental release and large-scale use

READINESS / STATUS LIMIT

- SBCC + DLC → state and district monitoring
- RDAC → recombinant-DNA advice

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: SBCC + DLC → state and district monitoring.

07

STAGE 07

LAW AND REGULATOR SPLIT

LAW AND REGULATOR SPLIT · RULES 1989 · CORE GMO BIOSAFETY FRAMEWORK · CONTAINED WORK · ENVIRONMENTAL APPRAISAL · GM-FOOD SAFETY APPROVAL AND LABELLING TRACK · CROP SYSTEM · AGRONOMIC EVALUATION AND VARIETY PROCESSES

SYSTEM / DEVICE / INSTITUTION	DECISIVE TECHNICAL DISTINCTION	STATUS / UNIT / EVIDENCE LIMIT
RULES 1989 → core GMO biosafety framework	GEAC → environmental appraisal	ICAR / CROP SYSTEM → agronomic evaluation and variety processes
RCGM → research / contained work	FSSAI → GM-food safety approval and labelling track	RULES 1989 → core GMO biosafety framework

SYSTEM / DEVICE / INSTITUTION

- RULES 1989 → core GMO biosafety framework
- RCGM → research / contained work

DECISIVE TECHNICAL DISTINCTION

- GEAC → environmental appraisal
- FSSAI → GM-food safety approval and labelling track

STATUS / UNIT / EVIDENCE LIMIT

- ICAR / CROP SYSTEM → agronomic evaluation and variety processes
- RULES 1989 → core GMO biosafety framework

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: ICAR / CROP SYSTEM → agronomic evaluation and variety processes.

08

STAGE 08

INDIAN DECISION TIMELINE

INDIAN DECISION TIMELINE · 09 FEB 2010 · BT BRINJAL MORATORIUM · 30 MAR 2022 · SDN-1/2 EXEMPTION OM · 17 MAY 2022 · DBT GENOME-EDITED PLANT GUIDELINES · 23 JUL 2024

09 FEB 2010 → Bt brinjal moratorium → 30 MAR 2022 → SDN-1/2 exemption OM → 17 MAY 2022 → DBT genome-edited plant guidelines → 23 JUL 2024 → Supreme Court DMH-11 split verdict → 04 APR 2025 → two genome-edited rice varieties announced

CONCEPT / ARCHITECTURE

- 09 FEB 2010 → Bt brinjal moratorium
- 30 MAR 2022 → SDN-1/2 exemption OM

MECHANISM / APPLICATION

- 17 MAY 2022 → DBT genome-edited plant guidelines
- 23 JUL 2024 → Supreme Court DMH-11 split verdict

READINESS / STATUS LIMIT

- 04 APR 2025 → two genome-edited rice varieties announced
- 09 FEB 2010 → Bt brinjal moratorium

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: 23 JUL 2024 → Supreme Court DMH-11 split verdict.

09

STAGE 09

BIOSAFETY MECHANISM MAP

BIOSAFETY MECHANISM MAP · TARGET PEST PRESSURE · RESISTANCE EVOLUTION · SEED MOVEMENT · GENE-FLOW EXPOSURE · TRAIT EXPRESSION · NON-TARGET EXPOSURE PATHWAY · SIMILAR DNA SITES

TARGET PEST PRESSURE → resistance evolution

POLLEN / SEED MOVEMENT → gene-flow exposure → TRAIT EXPRESSION → non-target exposure pathway → SIMILAR DNA SITES → possible off-target edit

STARTING CONDITION / INPUT

- TARGET PEST PRESSURE → resistance evolution
- POLLEN / SEED MOVEMENT → gene-flow exposure

SCIENTIFIC OR ENGINEERING MECHANISM

- TRAIT EXPRESSION → non-target exposure pathway
- SIMILAR DNA SITES → possible off-target edit

OUTPUT / FAILURE BOUNDARY

- RESPONSE → characterisation, mitigation and monitoring
- TARGET PEST PRESSURE → resistance evolution

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: SIMILAR DNA SITES → possible off-target edit; RESPONSE → characterisation, mitigation and monitoring.

10

STAGE 10

FARMER-LIVELIHOOD EVALUATION

FARMER-LIVELIHOOD EVALUATION · INTENDED AGRONOMIC CONSTRAINT · FIELD PERFORMANCE · YIELD STABILITY AND INPUT EFFECT · FARM ECONOMICS · SEED COST, LABOUR, PESTICIDE AND RISK · EXTENSION, COMPETITION, STEWARDSHIP AND MARKETS · CONTEXT-SPECIFIC WELFARE, NOT TECHNIQUE DETERMINISM

TRAIT → intended agronomic constraint

FIELD PERFORMANCE → yield stability and input effect → FARM ECONOMICS → seed cost, labour, pesticide and risk → INSTITUTIONS → extension, competition, stewardship and markets

CONCEPT / ARCHITECTURE

- TRAIT → intended agronomic constraint
- FIELD PERFORMANCE → yield stability and input effect

MECHANISM / APPLICATION

- FARM ECONOMICS → seed cost, labour, pesticide and risk
- INSTITUTIONS → extension, competition, stewardship and markets

READINESS / STATUS LIMIT

- VERDICT → context-specific welfare, not technique determinism
- TRAIT → intended agronomic constraint

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: FARM ECONOMICS → seed cost, labour, pesticide and risk.

11

STAGE 11

APPROVAL AND CLAIM FIREWALL

APPROVAL AND CLAIM FIREWALL · LAB RESULT · BOUNDED MOLECULAR PROPOSITION · FIELD PERMISSION · TRIAL STAGE · FOOD APPROVAL · SEPARATE REGULATORS · VARIETY RELEASE

LAB RESULT → bounded molecular proposition → CONTAINED / FIELD PERMISSION → trial stage → ENVIRONMENTAL / FOOD APPROVAL → separate regulators → VARIETY RELEASE / SEED ACCESS / CULTIVATION → separate evidence → POST-RELEASE MONITORING → resistance, ecology and farmer outcomes

SYSTEM / DEVICE / INSTITUTION

- LAB RESULT → bounded molecular proposition
- CONTAINED / FIELD PERMISSION → trial stage

DECISIVE TECHNICAL DISTINCTION

- ENVIRONMENTAL / FOOD APPROVAL → separate regulators
- VARIETY RELEASE / SEED ACCESS / CULTIVATION → separate evidence

STATUS / UNIT / EVIDENCE LIMIT

- POST-RELEASE MONITORING → resistance, ecology and farmer outcomes
- LAB RESULT → bounded molecular proposition

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: POST-RELEASE MONITORING → resistance, ecology and farmer outcomes.

E

EXTRA

SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT · INNOVATION-SPEED DEBATE · LIGHTER TREATMENT FOR SDN-1/SDN-2 CAN ACCELERATE TRAIT DEVELOPMENT AND · BIOSAFETY DEBATE · CRITICS ARGUE THAT EVEN PRECISE EDITS MAY HAVE ECOLOGICAL · PUBLIC-TRUST DEBATE · REGULATORY DIFFERENTIATION IS SCIENTIFICALLY MEANINGFUL, BUT PUBLIC COMMUNICATION CAN · TRADE AND TRACEABILITY DEBATE

OPTIONAL SCIENTIFIC / ENGINEERING DEPTH

- CAUTION: Innovation-speed debate: lighter treatment for SDN-1/SDN-2 can accelerate trait development and public-sector crop improvement.
- CAUTION: Biosafety debate: critics argue that even precise edits may have ecological or off-target consequences needing robust oversight.
- CAUTION: Public-trust debate: regulatory differentiation is scientifically meaningful, but public communication can lag behind.
- CAUTION: Trade and traceability debate: export markets may classify or monitor edited crops differently.

READINESS, UNIT OR STATUS LIMIT

- CAUTION: Federal/agricultural politics: crop adoption interacts with farmer trust, seed systems and state-level political acceptance.
- CAUTION: Gene editing could matter for climate resilience, disease resistance, nutritional traits and public-sector breeding.
- CAUTION: GM and gene-editing capacity also reflect India's ability to convert molecular biology into agricultural technology.
- CAUTION: A nuanced regulatory model can support innovation without erasing biosafety review.

COMPARATIVE TECHNOLOGY / GOVERNANCE NUANCE

- CAUTION: Ecological uncertainty: gene flow, pollinator concerns, resistance development and biodiversity effects remain important.
- CAUTION: Institutional complexity: research oversight, biosafety testing, environmental release and agricultural release involve different bodies.
- CAUTION: Public communication gap: technical distinctions between transgenic and gene-edited products are not easy to communicate in politics.
- CAUTION: Legal uncertainty: later judicial or regulatory developments can affect rollout status, especially in contentious cases such as GM mustard.

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.

END OF MASTER CANVAS • poster embeds this exact master • tiled pages are overlapping crops of the same pixels